Intermediate Quantitative Methods

Regression anatomy

Kevin Munger

Week 2, part 2

European University Institute, SPS

Today

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- What least squares does

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- What “holding constant” means

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- Interactions

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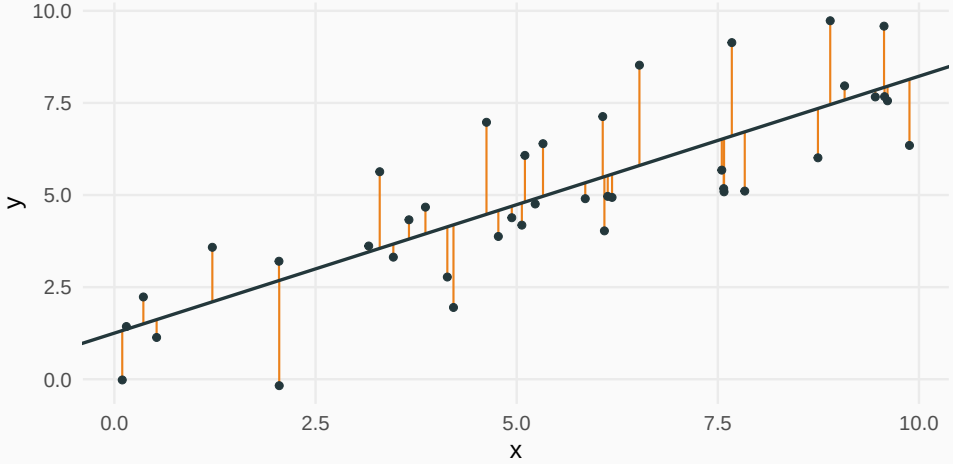
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- Residual: $e_i = y_i - \hat{y}_i$, the vertical distance to the line

A fitted line and its residuals, $n = 40$



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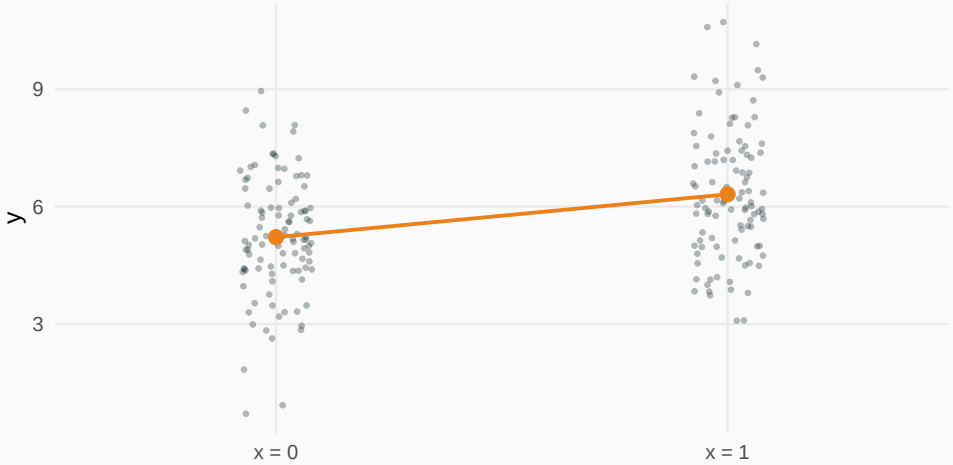
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- So residuals cannot tell you whether x is confounded

A binary predictor, 100 per group



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- Monday's lab does all of this in R

Holding constant

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- Where no units share the same x_2, the model fills the gap by assuming linearity

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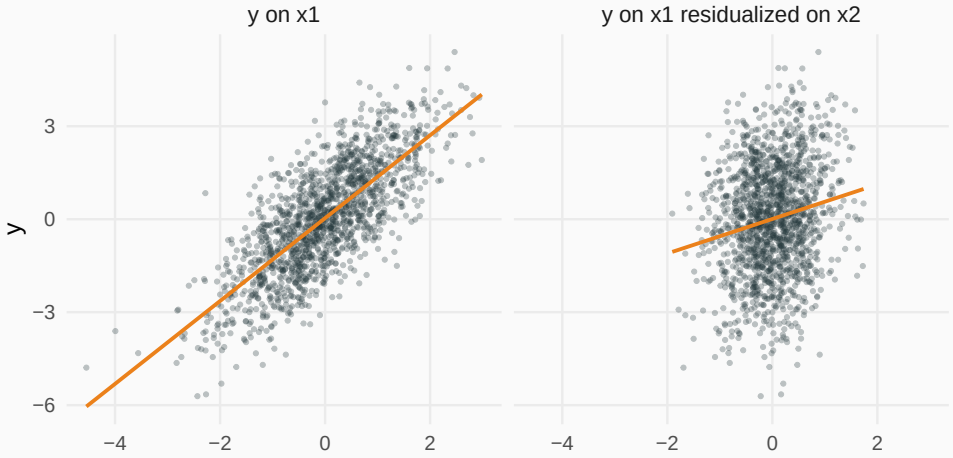
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- Frisch–Waugh–Lovell theorem

Simulated data, true $\beta_1 = 0.5$, $n = 1{,}500$



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- Every added control narrows the comparison being made

Uncertainty in a slope

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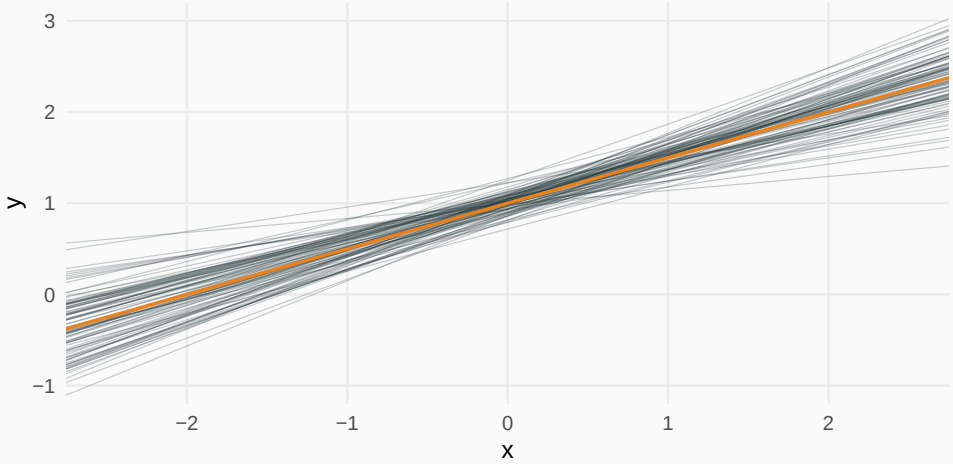
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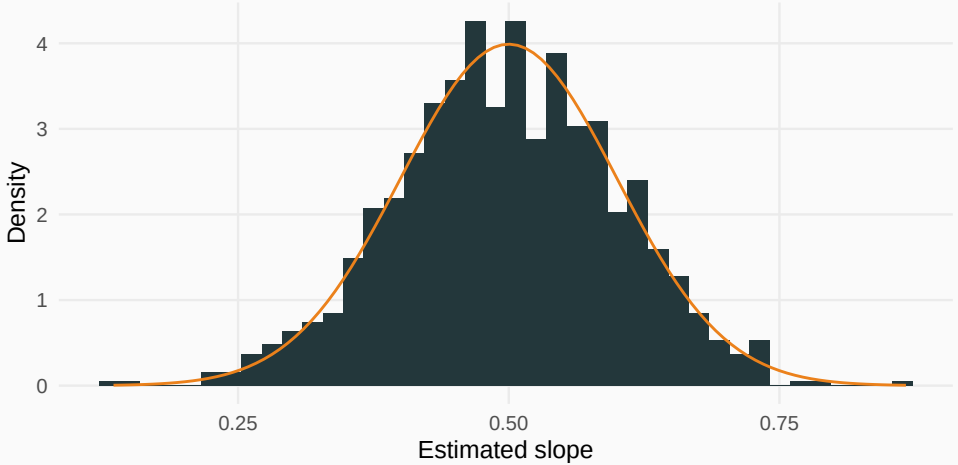
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- A different sample gives a different line
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- Simulation: true slope 0.5, $n = 100$, 1,000 datasets

Fitted lines from 100 simulated datasets



Slopes from 1,000 simulated datasets



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- Then as last week: $t = \hat{\beta}/\widehat{SE}$, 95% CI $= \hat{\beta} \pm 1.96 \times \widehat{SE}$

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- No software output checks the first two

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- Coverage: do 95% intervals contain 0.5 in 95% of datasets?

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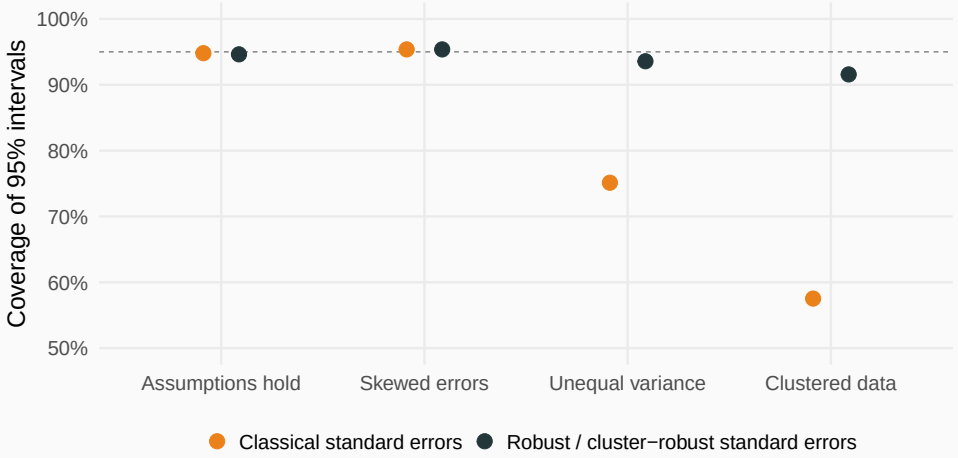
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- Clustered data: 20 groups of 10, x and part of the error shared within group

Coverage of 95% intervals, 4,000 datasets per scenario



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- 20 clusters is few: the correction itself is approximate

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- Lab 2: clustered and bootstrap standard errors on the ESS

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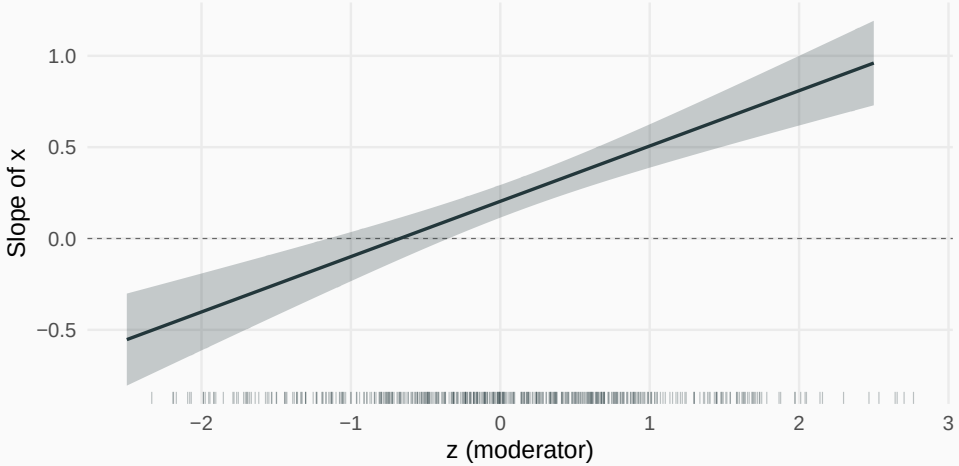
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- The slope of x is no longer one number: $\beta_1 + \beta_3 z$
- z: the moderator
- β_1 alone: the slope of x when $z = 0$, nothing more

Slope of x across z, with 95% interval



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- Check the interaction is linear and that data exist across z (Hainmueller, Mummolo & Xu 2019)

What to take from this

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- Worry about validity and representativeness before normality
- When in doubt about a method, simulate it

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Questions